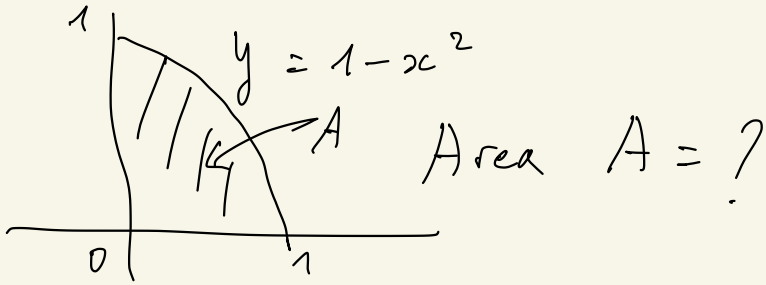


Reminder about last lecture.

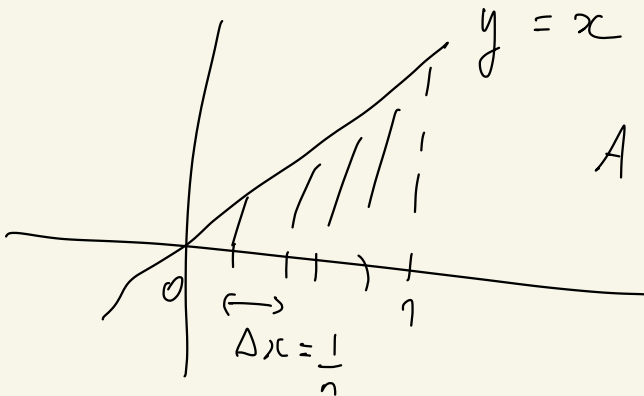
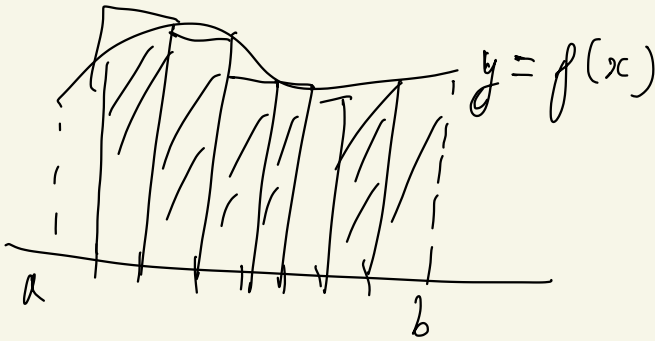
Calculus A(1)

2022-11-28

Areas below graphs



Approximation by small rectangles.



$$A = \frac{1 \cdot 1}{2} = \frac{1}{2}$$

Let us recover this area $A = \frac{1}{2}$
using the rectangles method.

$$f(x) = x$$

$$S = \sum_{k=0}^{n-1} \Delta x f\left(\frac{k}{n}\right) = \sum_{k=0}^{n-1} \frac{k}{n} \cdot \frac{1}{n}$$

$$= \sum_{k=0}^{n-1} \frac{k}{n^2}$$

$$= \frac{1}{n^2} \sum_{k=0}^{n-1} k$$

Recall: $\sum_{k=0}^{n-1} k = \frac{n(n-1)}{2}$

Using this formula

$$S = \frac{1}{n^2} \frac{n(n-1)}{2} = \frac{n-1}{2n} \xrightarrow{n \rightarrow +\infty} \frac{1}{2} = A.$$

Other trick to compute

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Trick: $k^2 - (k-1)^2 = 2k - 1$

We get :

$$\sum_{k=1}^n (2k-1) = \sum_{k=1}^n k^2 - (k-1)^2$$

$$LHS = RHS$$

$$\begin{aligned} \text{RHS: } & \cancel{(1^2 - 0^2)} + \cancel{(2^2 - 1^2)} + \cancel{(3^2 - 2^2)} \\ & \quad \quad \quad + \dots + \cancel{((n-1)^2 - (n-2)^2)} \\ & \quad \quad \quad + n^2 - \cancel{(n-1)^2} \\ = & n^2 - 0^2 = n^2 \end{aligned}$$

RHS is what we call a telescoping sum.

$$S_0 \text{ RHS} = n^2$$

$$\text{LHS:} \quad 2 \sum_{k=1}^n k - \underbrace{\sum_{k=1}^n 1}_n$$

$$\text{So} \quad 2 \sum_{k=1}^n k - n = n^2$$

$$\text{So:} \quad \sum_{k=1}^n k = \frac{n^2 + n}{2} = \frac{n(n+1)}{2}$$

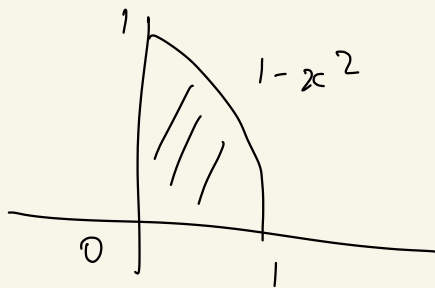
Philosophy: analogue to

$$\frac{d}{dx} (x^2) = 2x$$

$$\downarrow$$

$$k^2 - (k-1)^2 = 2k - 1$$

Back to



$$f(x) = 1 - x^2$$

$$S = \sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) \cdot \Delta x$$

where: $\Delta x = \frac{1}{n}$ (upper rectangles)

$$S = \sum_{k=0}^{n-1} \left(1 - \left(\frac{k}{n}\right)^2\right) \cdot \frac{1}{n}$$

$$= \underbrace{\frac{1}{n} \left(\sum_{k=0}^{n-1} 1 \right)}_1 - \underbrace{\frac{1}{n^3} \sum_{k=0}^{n-1} k^2}_{=?}$$

$$S = 1 - \frac{1}{n^3} \sum_{k=0}^{n-1} k^2 = ?$$

One can again find a "closed formula" for S .

Formula : $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

One proof is by induction.

One can also prove it by telescoping

$$\frac{d}{dx}(x^3) = 3x^2$$

$$k^3 - (k-1)^3 \iff 3k^2$$

$$\begin{aligned} k^3 - (k-1)^3 &= k^3 - (k^3 - 3k^2 + 3k - 1) \\ &= 3k^2 - 3k + 1 \end{aligned}$$

$$\text{So } \sum_{k=1}^n k^3 - (k-1)^3 = \sum_{k=1}^n (3k^2 - 3k + 1)$$

LHS = telescoping
Sum.

$$\begin{aligned} &= 3 \left(\sum_{k=1}^n k^2 \right) - 3 \sum_{k=1}^n k \\ &\quad + \sum_{k=1}^n 1 \end{aligned}$$

$$= 3 \left(\sum_{k=1}^n k^2 \right) - 3 \frac{n(n+1)}{2} + n$$

what we
want to compute

$$\sum_{k=1}^n k^3 - (k-1)^3 = (\cancel{1^3} - \cancel{0^3}) + (\cancel{2^3} - \cancel{1^3}) + (\cancel{3^3} - \cancel{2^3}) + \dots + (\cancel{n^3} - \cancel{(n-1)^3})$$

$$= n^3 - 0^3 = n^3$$

Conclusion :

$$3 \left(\sum_{k=1}^n k^2 \right) - 3 \frac{n(n+1)}{2} + n = n^3$$

$$\Rightarrow \sum_{k=1}^n k^2 = \frac{1}{3} \left(n^3 + \frac{3n(n+1)}{2} - n \right)$$

$$= \frac{n(n+1)(2n+1)}{6}$$

Back to

$$S = 1 - \frac{1}{n^3} \sum_{k=1}^{n-1} k^2$$

$$= 1 - \frac{1}{n^3} \frac{(n-1)n(2n-1)}{6}$$

$$\xrightarrow{n \rightarrow \infty} 1 - \frac{2}{6} = 1 - \frac{1}{3} = \frac{2}{3}$$

Conclusion : Area = $\frac{2}{3}$

(can do the same computation for lower rectangles)

Riemann sums

Riemann sums are a generalization of the above computations to general functions.

$$f: [a, b] \longrightarrow \mathbb{R}$$

(any function, f can take both positive and negative values)

We subdivide $[a, b]$ into n subintervals, i.e. we choose

$n-1$ points $x_1, \dots, x_{n-1}$ s.t.

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n$$



Let $P = \{x_0, x_1, \dots, x_{n-1}, x_n\}$

where $\begin{cases} x_0 = a \\ x_n = b \end{cases}$

P is called a partition of $[a, b]$
 $[a, b]$ is divided into n subintervals

$\underset{\substack{\parallel \\ a}}{[x_0, x_1]}, [x_1, x_2], \dots, [x_{n-1}, \underset{\substack{\parallel \\ b}}{x_n}]$

k^{th} subinterval is $[x_{k-1}, x_k]$

for $k \in \{1, 2, \dots, n\}$.

The width (or length) of

the k^{th} subinterval is

$$\Delta x_k = x_k - x_{k-1}$$

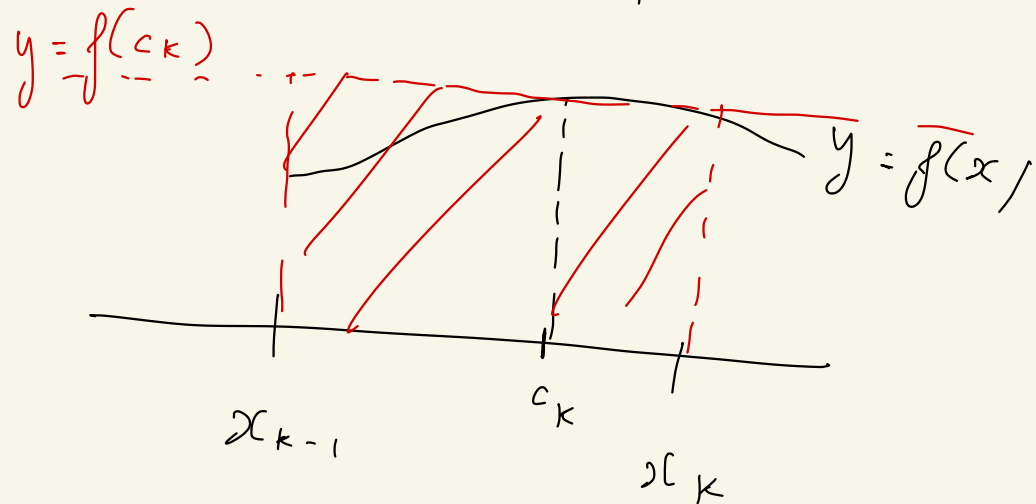
If all the subintervals have the

same length, we have

$$\Delta x_k = \frac{b-a}{n} \quad (\forall k)$$

We do not assume that for general partitions.

For each $k \in \{1, \dots, n\}$,
select some point $c_k \in [x_{k-1}, x_k]$



The choice of c_k defines a
rectangle whose height is
 $|f(c_k)|$

Area of the rectangle

$$= \Delta x_k \cdot |f(c_k)|$$

Finally :

let $S_P := \sum_{k=1}^n f(c_k) \Delta x_k$
Partition of $[a, b] \rightarrow P$

= Riemann sum of
 f on $[a, b]$ associated
to the partition P .

Idea : When P becomes "finer"

S_P should (in "nice" cases)
Converge to some value
(which is the "area" if $f \geq 0$)

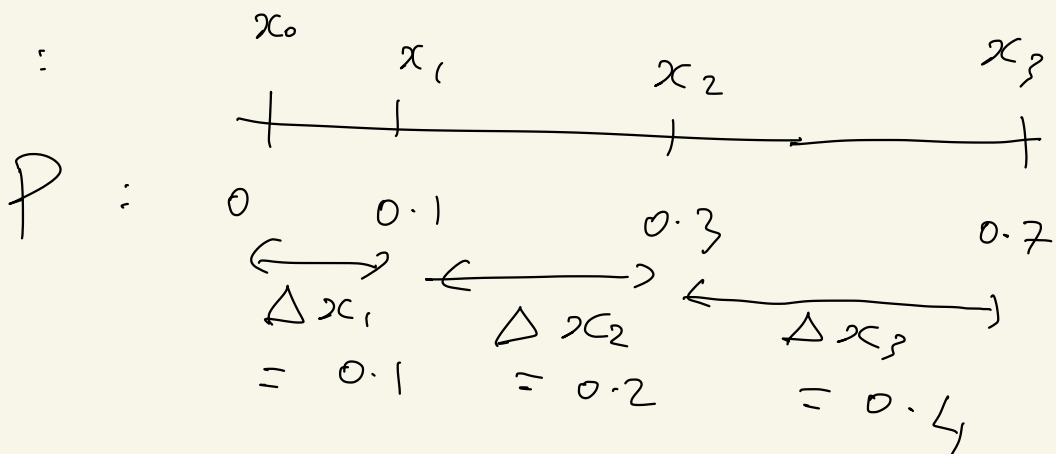
Let $\|P\| := \max_{k \in \{1, \dots, n\}} \Delta x_k$

↑
"Norm of P"

Norm of P = max of length of sub intervals in P.

The smaller $\|P\|$, the "finer" P is.

Ex :



$$\|P\| = 0.4$$

Question : $\lim_{\|P\| \rightarrow 0} S_P = ?$

What does this mean?

Def : let $f: [a, b] \rightarrow \mathbb{R}$
(no assumptions on f)

We say that f is integrable
on $[a, b]$ if $\lim_{\|P\| \rightarrow 0} S_P$ exists,

i.e. $\exists I \in \mathbb{R}$ s.t

$\forall \varepsilon > 0, \exists \delta > 0$ s.t if $\|P\| < \delta$

then $|S_P - I| < \varepsilon$

I is called the definite integral
of f over $[a, b]$

Leibniz notation : $\int_a^b$ ← upper limit of integration

I def

$\int_a^b f(x) dx$
↑
a ← lower limit of integration
↑
integral sign
↑
f(x) ← integrand
↑
x ← variable of integration

$\int$ represents a "S"
(for "sum")

dx is analogous to Δx
(but infinitesimal)

Rem : The name of the variable of integration does not matter

We could write $\int_a^b f(t) dt$

or $\int_a^b f(u) du$

or ...

x = dummy variable

Questions :

① How do we prove a function is integrable?

(difficulty: there are so many different Riemann sums.)

② How do we compute $\int_a^b f(x) dx$ if it exists?

For ①, we have the following result:

Theorem : Let $f: [a, b] \rightarrow \mathbb{R}$

If f is C^0 on $[a, b]$ then

f is integrable on $[a, b]$,
ie $\int_a^b f(x) dx$ exists.

Idea of the proof :

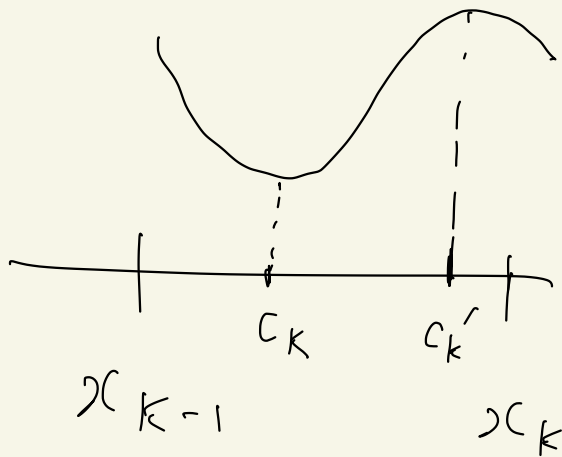
Let $P = \{ \underset{\substack{\parallel \\ a}}{x_0}, x_1, \dots, x_{n-1}, \underset{\substack{\parallel \\ b}}{x_n} \}$

be a partition of $[a, b]$.

Let $c_k \in [x_{k-1}, x_k]$ s.t.
 f has a global min at $x=c_k$

on $[x_{k-1}, x_k]$

Similarly, let $c'_k \in [x_{k-1}, x_k]$
s.t. f has a global max
at $x = c'_k$.



$$\text{Let } U = \sum_{k=1}^n \Delta x_k \cdot f(c'_k)$$

$$\text{and } L = \sum_{k=1}^n \Delta x_k \cdot f(c_k)$$

We have $L \leq U$

because $f(c_k) \leq f(c_{k'})$
(and $\Delta x_k > 0$).

$U - L$ is arbitrarily small
if $\|P\|$ is small enough.

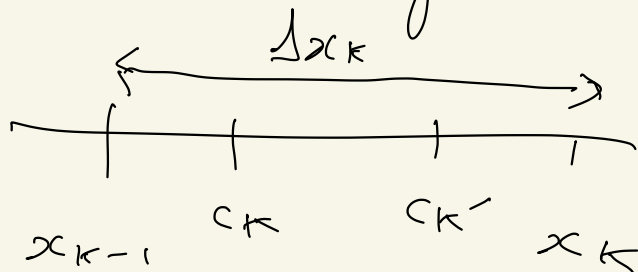
$$U - L = \sum_{k=1}^n \Delta x_k (f(c_{k'}) - f(c_k))$$

Claim: $\forall \varepsilon > 0$, if Δx_k
is small enough, $(\forall k)$ then

$$0 \leq f(c_{k'}) - f(c_k) < \varepsilon$$

To prove this claim, we need

to use the fact that f is C^0



The problem is: k is not fixed.

We want

$$\forall k, |\Delta x_k| < \delta \Rightarrow f(c_k) - f(c_k') < \varepsilon$$

$\Rightarrow$

We need "uniform continuity"

(see HW 7 of f for more details).

The previous result is only a sufficient condition for integrability. There exists non- C^0

functions which are integrable.

e.g. piecewise C^0 functions are integrable (finite number of discontinuities)

• Monotonic functions are integrable.

For a function to be integrable, the region between $y = f(x)$ and the x -axis has to be well-approximated by "thin" rectangles.

Example of non-integrable function

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

$$f: [0, 1] \longrightarrow \mathbb{R}$$

Claim: f is not integrable on $[0, 1]$.

Let $P = \left\{ \underset{\substack{\parallel \\ a}}{x_0}, x_1, x_2, \dots, x_n, \underset{\substack{\parallel \\ b}}{x_n} \right\}$

$\forall k \in \{1, \dots, n\},$

• $\exists c_k \in [x_{k-1}, x_k]$ s.t

$c_k \notin \mathbb{Q} \implies \text{Riemann sum}$

• $\exists c'_k \in [x_{k-1}, x_k]$ s.t

$c'_k \in \mathbb{Q}.$

$\implies \text{Riemann sum}$

$$L = \sum_{k=1}^n \underbrace{f(c_k)}_0 \Delta x_k = 0$$

$$U = \sum_{k=1}^n \underbrace{f(c_k')}_{=1} \Delta x_k$$

$$U = \sum_{k=1}^n \Delta x_k = \text{length of } [0, 1]$$

$$L = 0, \quad U = 1$$

$L \neq U$. P can have norm as small as we want

So $\lim_{\|P\| \rightarrow 0} S_P$ does not exist

and f is not integrable.

Basic properties of the definite integral

When we defined $\int_a^b f(x) dx$,

we moved from a to b P
from left to right

$$\Delta x_k = x_k - x_{k-1} > 0$$

If we move from b to a

$\bar{P}$ then $x_0 = b > x_1 > x_2 > \dots > x_n$
"a"

$$\Delta x_k = x_k - x_{k-1} < 0$$

$$S_{\bar{P}} = -S_P$$

(where $\bar{P}$: from b to a)

Motivated by that, we make the following definition

$$\left[\begin{array}{l} \text{Def : } \cdot a < b \\ \quad \cdot \text{ let } \int_a^b f(x) dx := - \int_b^a f(x) dx \\ \cdot \text{ if } a = b, \text{ let } \int_a^a f(x) := 0 \end{array} \right.$$

The definite integrals enjoy the following basic algebraic properties and inequalities:

Theorem : let $f, g : [a, b] \rightarrow \mathbb{R}$
integrables.

① (Sum rule)

$f + g$ is integrable on $[a, b]$

and
$$\int_a^b (f(x) + g(x)) dx$$

$$= \int_a^b f(x) dx + \int_a^b g(x) dx$$

② $\forall \lambda \in \mathbb{R}$, $\lambda \cdot f$ is integrable

and
$$\int_a^b \lambda \cdot f(x) dx = \lambda \cdot \int_a^b f(x) dx$$

(Additivity)

③ $\forall a_1, a_2, a_3 \in [a, b]$, we

have
$$\int_{a_1}^{a_3} f(x) dx = \int_{a_1}^{a_2} f(x) dx + \int_{a_2}^{a_3} f(x) dx$$

(4) (Min - Max inequality)

$$(b-a) \min_{[a,b]}(f) \leq \int_a^b f(x) dx \leq (b-a) \cdot \max_{[a,b]}(f)$$

(5) if $\forall x \in [a,b], f(x) \geq g(x)$

Then
$$\int_a^b f(x) dx \geq \int_a^b g(x) dx$$

Special case : if $g = 0$

if $\forall x \in [a,b], f(x) \geq 0$ then

$$\int_a^b f(x) dx \geq 0$$

In order to give a geometric interpretation of these inequalities,

let us make the following definition.

Def (Area under a curve)

Let $f : [a, b] \rightarrow \mathbb{R}$
integrable and s.t

$$\forall x \in [a, b], \quad f(x) \geq 0$$

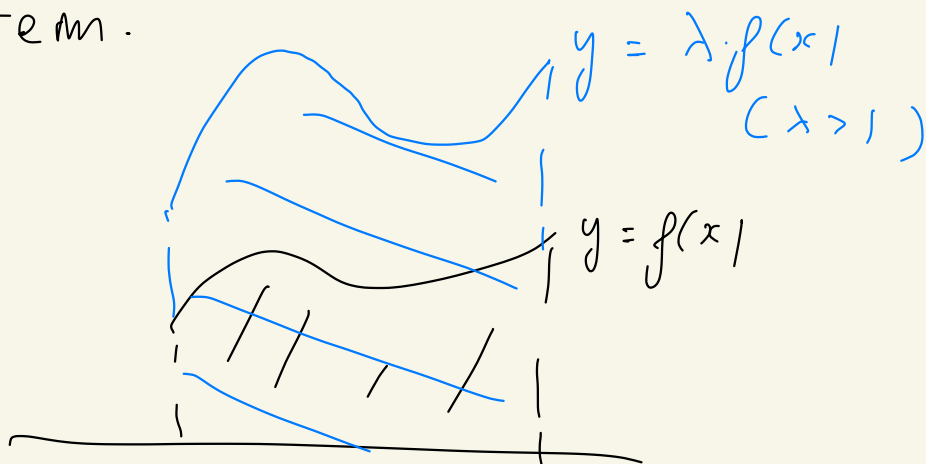
We define the area A
between the graph $y = f(x)$
and the x -axis to be

$$A := \int_a^b f(x) dx$$



We can now illustrate the theorem.

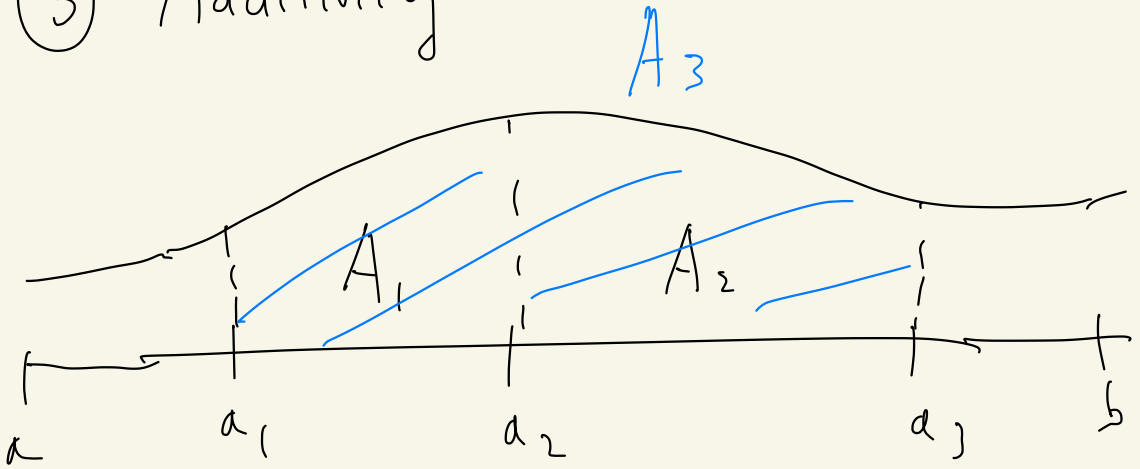
(2)



Blue area = $\lambda \cdot$ (black area)
because we make a vertical stretch by λ .

$$\int_a^b \lambda \cdot f(x) dx = \lambda \cdot \int_a^b f(x) dx$$

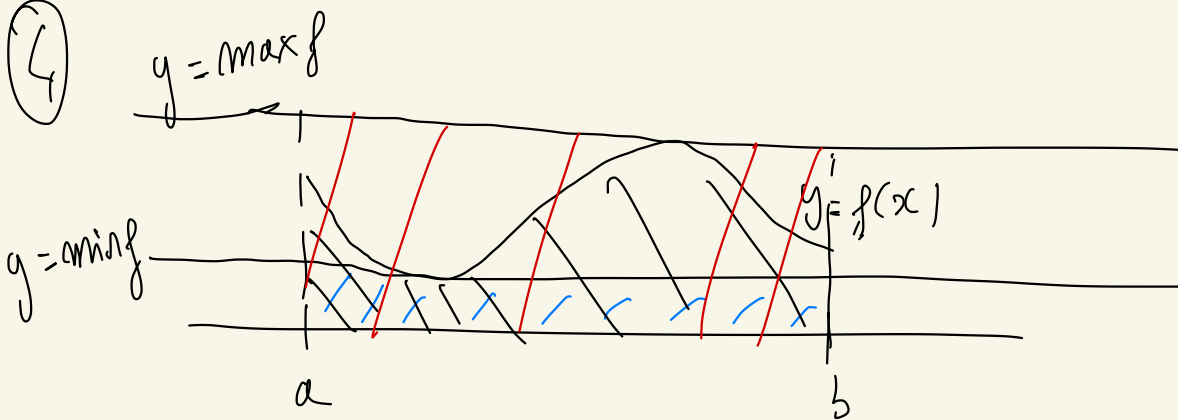
③ Additivity



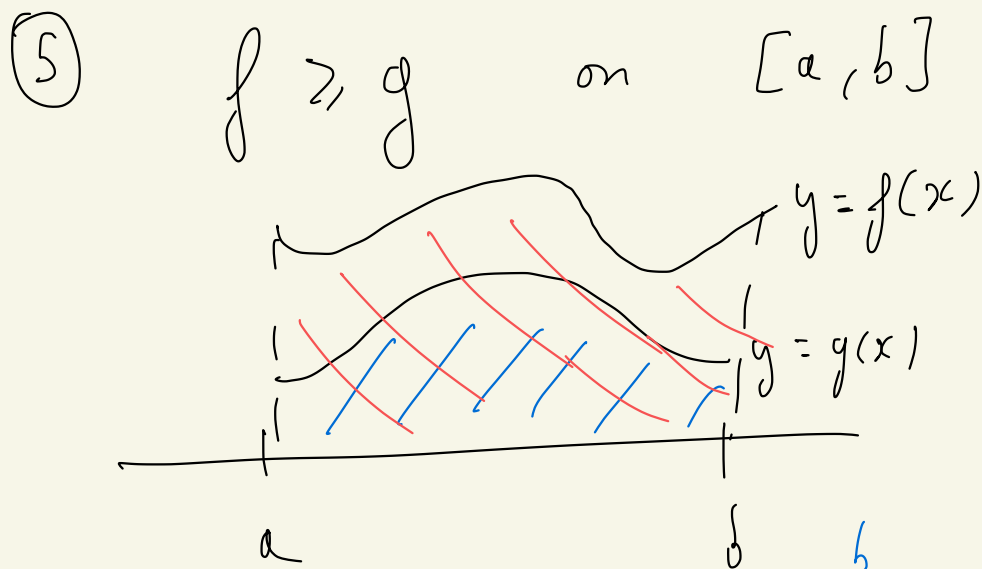
$$A_3 = A_1 + A_2$$

$$\begin{array}{ccc} \parallel & \parallel & \parallel \\ \int_{a_1}^{a_3} f(x) dx & \int_{a_1}^{a_2} f(x) dx & \int_{a_2}^{a_3} f(x) dx \end{array}$$

④



$$\begin{array}{ccc}
 \text{blue area} & \leq \int_a^b f(x) dx \leq & \text{red area} \\
 \parallel & & \parallel \\
 (b-a) \min_{[a,b]} f & & (b-a) \max_{[a,b]} f
 \end{array}$$



$$\int_a^b f(x) dx \geq \int_a^b g(x) dx$$

The proof of the theorem
uses the definition of the

definite integral in terms of
Riemann sums.

Ex of proof : (4)

Let P = a partition of $[a, b]$

Riemann sum
↓
 $P = \{ \underset{\substack{\text{"} \\ a}}{x_0}, x_1, \dots, \underset{\substack{\text{"} \\ b}}{x_n} \}$

$$S_P := \sum_{k=1}^n \Delta x_k f(c_k)$$

$$c_k \in [x_{k-1}, x_k]$$

$$S_P \leq \sum_{k=1}^n \Delta x_k \cdot \max_{[a, b]}(f)$$

because $\forall k, f(c_k) \leq \max_{[a,b]} f$
and $\Delta x_k \geq 0$.

$$= \max_{[a,b]} f \underbrace{\sum_{k=1}^n \Delta x_k}_{b-a}$$

So $\forall$ P partition
 $\forall$ choice of c_k

We got $S_p \leq \max_{[a,b]} f \cdot (b-a)$

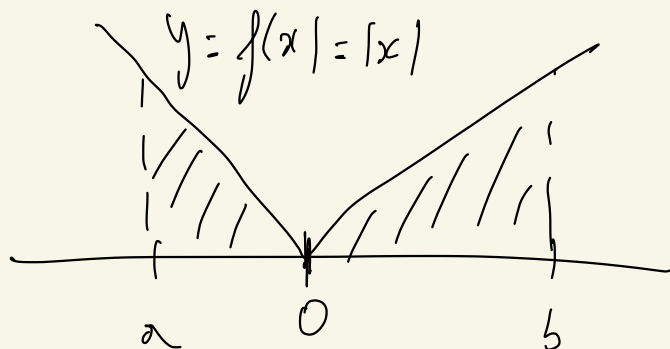
In the limit, we get-

$$\int_a^b f(x) dx \leq \max_{[a,b]} f \cdot (b-a)$$

Ex : Assume $a < 0 < b$

Q : $\int_a^b |x| dx = ?$

$\in C^0$ so integrable



$$\begin{aligned} \int_a^b |x| dx &= \int_a^0 |x| dx + \int_0^b |x| dx \\ &= \int_0^a x dx - \int_a^0 x dx \end{aligned}$$

$$= \frac{b^2}{2} + \frac{a^2}{2}$$